Unsupervised Domain Adaptive 3-D Detection With Data Adaption From LiDAR Point Cloud

Diankun Zhang, Xueqing Wang, Zhijie Zheng, and Xiaojun Liu, *Member, IEEE*

***Abstract*— Existing unsupervised domain adaptive (UDA) 3-D detection methods only address the domain gap caused by the prior size of 3-D bounding boxes between different datasets, which ignore the difference in the distribution of point clouds. To address this challenge, we propose an unsupervised domain adaptive 3-D detection by data adaption, which trains the model by transferring the source domain instances into the target domain scenes by adaptive point distribution. First, an instance transferring method is proposed for selecting and transferring suitable instances from the source domain into the target domain scene; Second, we propose an adaptive downsampling method to adjust the point cloud distribution of the transferred instances to approximate the points' distribution of the target domain. Finally, our method trains the randomly initialized detector with the pseudoinstances in the target domain. To the best of our knowledge, we first address the UDA problem of the 3-D detectors from the perspective of data. Extensive experiments on several popular datasets show that the proposed method outperforms the existing state-of-the-art (SOTA) methods by a large margin. Further experiments also show our approach is detector-agnostic and achieves consistent and significant gains on all types of 3-D detectors.**

***Index Terms*— 3-D detection, data-centric, point cloud, unsupervised domain adaptation (UDA).**

I. Introduction

LIGHT detection and ranging (LiDAR)-based 3-D object detection has developed rapidly in recent years, and it can be applied to many applications such as autonomous driving and robotics. Although LiDAR-based 3-D object detection benefits from the progress of image-based 2-D object detection, there are significant differences between them. The points in point clouds are generally unordered, sparse, and unevenly distributed, which make the convolutional neural network (CNN) not be applied to these kind of data directly. Therefore, LiDAR-based 3-D object detection has more diverse models for effective feature learning that consider the data format and characteristics of point clouds. Many 3-D object detection datasets have been built with larger and larger scales [1], [2], [3], which greatly promotes the development of deep learning methods [4], [5], [6], [7], [8], [9], [10], [11], [12], [13], [14], [15].

However, deep learning methods trained on one dataset (i.e., source domain) usually suffer severe performance degradation when they are evaluated on another dataset (i.e., target domain). For example, a 3-D detector trained on the Waymo Open Dataset [3] (collected in the U.S. by Waymo LiDAR) faces a dramatic performance drop (over 45%) [16] when it is evaluated on the KITTI dataset (KITTI) [1] (captured in European by Velodyne LiDAR). Previous works show that the scale mismatch problem (e.g., the size of cars in the U.S. is larger than that in European) is the major contributor to the domain gap between different datasets [3], [16], [17]. The problem is unique to 3-D detection because the size of the 3-D bounding boxes is usually consistent in the same dataset, which results in a narrow Gaussian distribution of the bounding box size [16], [18]. In contrast, the 2-D bounding boxes tend to have a large variety of sizes resulting from the distance between the object and the camera in one dataset [19], [20]. Considering that the annotation costs of 3-D bounding boxes are much higher than that of 2-D bounding boxes, unsupervised domain adaptation (UDA) for 3-D detection becomes a key issue of deep-learning-based methods.

Most of the existing methods only solve the domain gap caused by the prior knowledge of the 3-D bounding box size [16], [17]. However, the domain gap between different datasets is caused by not only the difference in the prior size of the bounding box but also the difference in point distribution due to the different LIDAR and scenes [3]. Table I shows the significant differences in the LiDAR performance used by the different datasets. Although UDA has been widely studied in 2-D detection which uses the feature alignment techniques [21], [22], [23], [24], [25] to address domain gaps due to lighting, color, texture, and style, they are not applicable to 3-D detection because there is no such information in point clouds. In addition, due to the irregular characteristics of point clouds, voxelization is required in the preprocessing of the voxel-based methods [5], [7]. As a strong bias of the network, these preprocessing parameters cannot be changed by training of fine-tuning. This makes the existing 3-D UDA methods [16], [17], [26] need additional alignment between

Manuscript received 19 September 2022; revised 30 October 2022; accepted 1 December 2022. Date of publication 2 December 2022; date of current version 20 December 2022. This work was supported by the National Natural Science Foundation of China under Grant 61827803. *(Corresponding author: Xiaojun Liu.)*

Diankun Zhang, Xueqing Wang, and Zhijie Zheng are with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100190, China, also with the Key Laboratory of Electromagnetic Radiation and Sensing Technology, Chinese Academy of Sciences, Beijing 100190, China, and also with the School of Electronic, Electrical and Communication Engineering, University of Chinese Academy of Sciences, Beijing 100049, China (e-mail: zhangdiankun19@mails.ucas.edu.cn; wangxueqing20@mails.ucas.edu.cn; zhengzhijie19@mails.ucas.edu.cn).

Xiaojun Liu is with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100190, China, and also with the Key Laboratory of Electromagnetic Radiation and Sensing Technology, Chinese Academy of Sciences, Beijing 100190, China (e-mail: lxjdr@mail.ie.ac.cn).

Digital Object Identifier 10.1109/TGRS.2022.3226570

1558-0644 © 2022 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:32:53 UTC from IEEE Xplore. Restrictions apply.

TABLE I
OVERVIEW OF DATASETS AND CORRESPONDING LIDAR SENSORS

Dataset	Size	Top-LiDAR type	Beam Angle	Vertical angular resolution	Collecting from
KITTI [1]	14999	64-beam	$[-24^\circ, 4^\circ]$	0.4°	Europe
nuScenes [2]	34149	32-beam	$[-16^\circ, 11^\circ]$	1.33°	U.S.A & Singapore
Waymo [3]	192484	64-beam	$[-18^\circ, 2^\circ]$	min to 0.2°	U.S.A

these parameters (e.g., the detection range of the detector pretrained on the Waymo dataset is much larger than the KITTI required, which is unnecessary when it is adapted to KITTI).

Unlike the existing UDA methods that fine-tune the model pretrained in the source domain [16], [17], we first propose to solve the UDA problem from the data perspective. In this article, an unsupervised domain adaptive 3-D detection method with data adaption is proposed, in which the existing annotated data in the source domain, the parameter information of the target domain LiDAR, and prior geometric statistics of the target domain are leveraged to adapt the source domain instance points in a proper distribution to the target domain, i.e., synthetic scenes with pseudoinstances and pseudoannotations. In this work, we aim to make the point distribution of the pseudoinstances in the synthetic scene close to the data distribution of the target domain and use them to train any 3-D detectors without any pretraining on the source domain data. In summary, our main **contributions** are.

1) We first address the UDA problem in 3-D detection from the perspective of data rather than model training or designing, i.e., a data-centric method.
2) We propose a concise yet effective data adaption method, which leverages the annotations in the source domain and the scenes in the target domain to synthesize scenes with pseudoinstances and pseudoannotations. In our method, an instance transferring method is proposed for selecting and transferring suitable instances from the source domain into the target domain scene, Then, an adaptive downsampling method is presented to adaptively adjust the point cloud distribution of the transferred instances to approximate the points' distribution of the target domain.
3) The proposed data adaption method outperforms the existing state-of-the-art (SOTA) UDA methods by a large margin on each type of 3-D detector and popular public datasets.

II. RELATED WORKS

A. LiDAR-Based 3-D Detector

The LiDAR-based object detection aims to localize and classify 3-D objects, which remains a challenging task due to the sparsity and irregularity of point clouds. The existing LiDAR-based detectors can be roughly divided into three categories: point-based, grid-based, and point-voxel-based.

1) Point-Based Detectors: The point-based target detection method directly processes the original point clouds and learns geometric features from them. Given the disorderly nature of point clouds, the point-based methods often use PointNet [27], PointNet++ [28], or their variants as the basic module for aggregating local or global point features to ensure that the disorderly nature of the point cloud is preserved. PointR-CNN [11] is a popular point-based two-stage detector that generates proposals from the segmentation results, followed by high-quality bounding box regression head which aggregates the point features inside the proposal. VoteNet [29] is a one-stage point-based 3-D detector where deep Hough voting is leveraged to regress the instance centroid. Point downsampling is a common method in point-based detectors to improve the efficiency of networks. One common downsampling method is farthest point sampling (FPS), which achieves approximate uniform sampling in space. Yang [30] proposes a hybrid sampling method that performs FPS in both the Euclidean space and feature space. Zhang [31] proposes an instance-aware downsampling strategy that significantly improves the efficiency of the network while reducing the number of sampling points. Overall, the point-based methods are usually intuitive and straightforward, and the main bottlenecks are insufficient learning capacity and limited efficiency.

2) Voxel-Based Detectors: Voxel-based detectors first divide unstructured point clouds into regular voxel (or pillar) grids, followed by a convolution neural network, which is similar to the 2-D object detection frameworks [32], [33]. VoxelNet [4] uses conventional 3-D convolution to learn the geometric pattern from sparse voxels which leads to unnecessary computational costs. Yan [7] presents an efficient sparse 3-D convolution backbone by leveraging the 3-D submanifold sparse convolution [34], [35]. Lang [6] proposes to further simplify voxels to pillars and resulting in higher efficiency. Zheng [8] presents the confident IoU-aware network to extract spatial-semantic features for object detection. Voxel-region-CNN (RCNN) [5] uses the voxel set aggregation module to further exploit the volumetric information in the proposal for better refinement. Centerpoint [36] extends CenterNet [37] and CornerNet [38] to 3-D detection, achieving high performance with a simple structure. Overall, the voxel-based methods usually have good efficiency and detection performance due to the sparse convolution backbone.

3) Point-Voxel-Based Detectors: The point-voxel-based methods attempt to use the joint representation of points and voxels to learn stronger features from point clouds while avoiding the disadvantages of the point method (i.e., poor memory locality) and the voxel method (i.e., quantization loss). Point-voxel RCNN (PV-RCNN) [12] and PV-RCNN++ [13] use voxel backbone to generate proposals

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:32:53 UTC from IEEE Xplore. Restrictions apply.

efficiently, and then extract point features in the proposals for accurate refinement. Noh [39] proposes an efficient memory module to enhance point features and directly fuse voxels and point features for a better balance of efficiency and accuracy.

In our work, the proposed data adaption will be evaluated on point-based, voxel-based (pillar-based), and point-voxel-based detectors, to verify the effectiveness and the generality of the proposed method.

B. Point Cloud Domain Adaptation

Domain adaptation has been widely studied in 2-D image data. Typically, some previous studies use adversarial learning to align different domain feature distributions [21], [22], [40], [41], and some use translating images to mitigate pixel-level domain gaps [42], [43]. The self-training-based methods [44], [45], [46], [47] generate pseudolabels for the unlabeled target domain to supervise the pretrained models.

Although there have been many excellent studies for 3-D detection tasks in recent years, only a few methods have been proposed to solve the UDA problem for 3-D detection. Some recent works are proposed to address the domain shift in shape classification [48] and semantic segmentation [49], [50], [51]. Point domain adaption network (PointDAN) [48] is proposed to jointly align local and global features using discrepancy loss and adversarial training for point cloud classification. Yi [49] designs a sparse voxel completion network to perform point cloud completion for domain adaptive semantic segmentation. Statistical normalization (SN) and output transformation (OT) are proposed to normalize the object size of the source domain by leveraging the geometric statistic of the target domain to mitigate the domain gap caused by the size prior [16]. Multi-level consistency (MLC)-Net [17] uses a multilevel consistency to facilitate cross-domain transfer.

Studies have shown that the difference in performance between different point cloud data mainly comes from the mismatch of the size of objects in different datasets [16], but the different environments, such as different cities, will also degrade the performance of the detector [3]. To address this problem, we propose an instance transferring method to transfer the instances from the source domain to the target domain, i.e., pseudoinstance in the target domain. The previous study used CycleGan [52] to transfer annotated images between domains due to the difference in light, style, and texture between different domains [53]. However, such information is not available in point clouds, and there are domain gaps caused by point distribution differences. Ground-truth (GT)-augmentation (AUG) [7] is a method of transferring 3-D instances in the same dataset, which can only transfer instances in fixed locations with fixed distribution and is not suitable for cross-domain instance transfer. We present an adaptive downsampling method to alleviate the distribution difference between the transferred instance and the target domain.

III. Data Adaption Method

A. Overview

We aim to transfer the instances in the source domain into the target domain scenes with proper distribution, i.e., a synthetic scene of the target domain containing pseudoinstances and pseudoannotations. Our method makes the distribution of the synthetic scenes close to the real annotated data in the target domain. The synthetic scenes are used to train random-initialized detectors with supervision, instead of the real annotated data in the target domain, to achieve the detection capability on the target domain.

As shown in Fig. 1, to achieve reasonable transfer between different domains (i.e., different scenes and different LiDAR), our data adaption is divided into two main parts: 1) transferring the suitable instances in the source domain to the target domain and 2) downsampling the transferred instances adaptively to fit the distribution properties of the target domain scene. Finally, the synthetic scenes with pseudoinstances will be used to train the detector.

B. Problem Definition

To facilitate the description of the problem, here we take a single point cloud $\mathcal{P}_s = \{\mathbf{p}_1^s, \mathbf{p}_2^s, \ldots\}$ and its corresponding bounding box set $\mathcal{B}_i = \{\mathbf{b}_1, \mathbf{b}_2, \ldots\}$ from the source domain as an example, where $\mathbf{p}_i^s$ is a point with coordinates (x_p^s, y_p^s, z_p^s) and $\mathbf{b}_i$ is a bounding box which contained position (x, y, z), size (l, w, h) and orientation β. Meanwhile, we have a point cloud $\mathcal{P}_t = \{\mathbf{p}_1^t, \mathbf{p}_2^t, \ldots\}$ where $\mathbf{p}_i^t$ has the same meaning with $\mathbf{p}_i^s$ but from the target domain. To simplify the problem and intuitively measure the effect of the pseudoinstances generated by the proposed data adaption method for network training, we assume that the target domain scene $\mathcal{P}_t$ does not contain any instance of interest. To train the detector by the pseudoinstances in the target domain scene, i.e., synthesize a point cloud scenes $\mathcal{P}_a$ with pseudobounding boxes $\mathcal{B}_s$, we leverage the instances indicated by B_i in the source domain and point could scene P_t in the target domain, as well as the information of the LiDAR and geometric statistics about the real instances in the target domain.

C. Instance Transferring Method

The instance transferring method aims to select and transfer suitable instances from the source domain into the target domain scene according to the pseudobounding boxes. The architecture is shown in Fig. 1. First, pseudobounding boxes are generated in the scene based on the a priori geometric statistics of the target domain; then, the matching instances in the source domain are selected to fill these pseudobounding boxes, based on their positions and yaw angles.

1) Generating Pseudobounding Boxes: First, we need to generate pseudobounding boxes to determine the position, size, and orientation of the pseudoinstance. The pseudobounding box $\mathbf{b}_p$ can be expressed as (1)

$$\mathbf{b}_p = \begin{bmatrix} x_p & y_p & z_p & l_p & w_p & h_p & \beta_p \end{bmatrix} \tag{1}$$

where x_p, y_p, z_p are the position of $\mathbf{b}_p$ on the X-axis, Y-axis, and Z-axis, l_p, w_p, h_p are the length, width, and height of $\mathbf{b}_p$, and β_p is the orientation (yaw) of $\mathbf{b}_p$.

Unlike the long-tail distribution commonly seen in the 2-D detection datasets, 3-D objects are relatively consistent in size and mainly follow the Gaussian distribution [16], [18]. We use a 3-D Gaussian distribution to describe the length l, width w, and height h of the pseudobounding boxes, the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:32:53 UTC from IEEE Xplore. Restrictions apply.

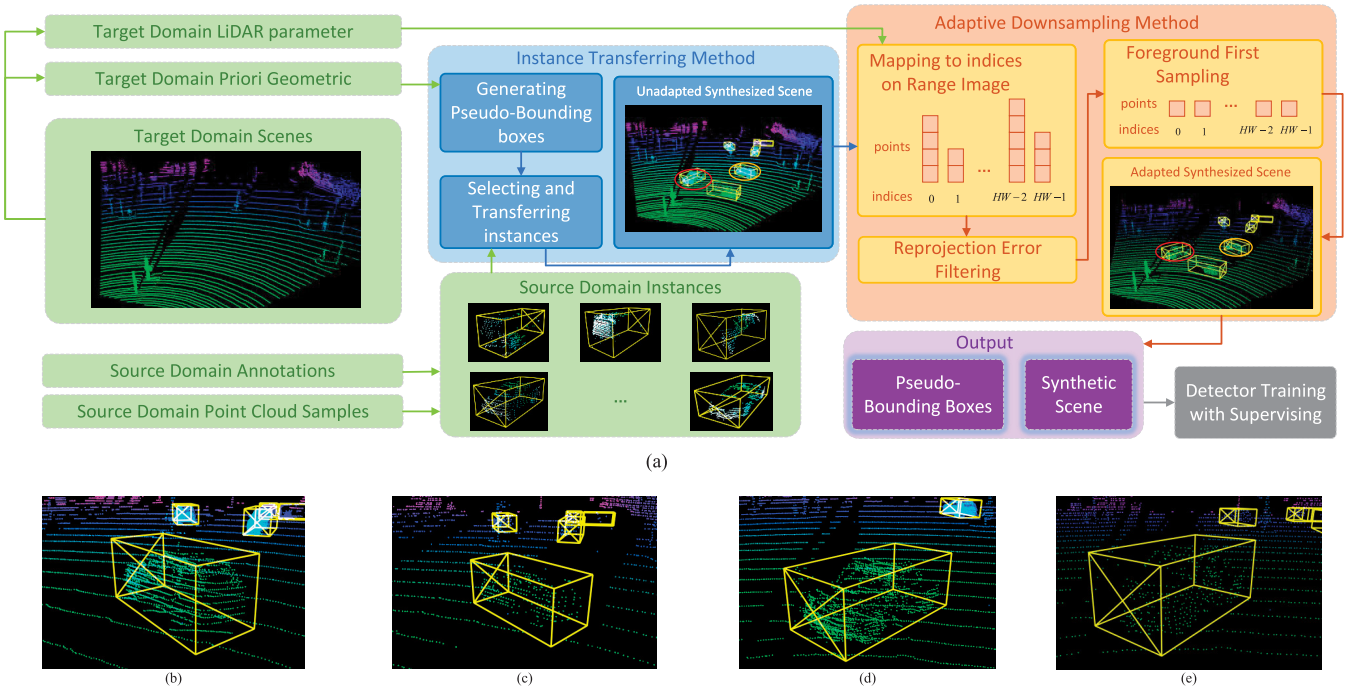


Fig. 1. (a) Data-adaptive framework consists of two phases: 1) selecting and transferring proper instances according to the pseudobounding boxes in the target domain by the instance transferring method and 2) generate pseudoinstances with approximate distribution to the target domain by adaptive downsampling method. (b) Detail of the instance marked by yellow circle in unadapted synthesized scene. (c) Detail of the instance marked by yellow circle in adapted synthesized scene. (d) Detail of the instance marked by red circle in unadapted synthesized scene. (e) Detail of the instance marked by red circle in adapted synthesized scene.

parameters of which are decided by the geometric statistics of the target domain. Since the instance can be anywhere on the plane, the horizontal position x, y and the orientation β of the pseudobounding box can be described as a uniform distribution. On the contrary, the vertical position z needs to ensure that the pseudoinstance is nearly on the ground, which can be calculated by the ground estimation of the target domain or determined according to the installation height of the LiDAR from the ground. In summary, the variables in (1) can be sampled from the following distributions, respectively:

$$\begin{aligned} x_p &\sim \mathcal{U}(x_{\min}, x_{\max}), \quad y_p \sim \mathcal{U}(y_{\min}, y_{\max}) \\ z_p &= f_r(x_p, y_p), \quad \theta_p \sim \mathcal{U}(-\pi, \pi) \\ (l_p, w_p, h_p) &\sim \mathcal{N}_3(\mathbf{x}|\mu_t, \mathbf{\Sigma}_t) \end{aligned} \tag{2}$$

where $x_{\min}, x_{\max}, y_{\min}, y_{\max}$ represents the horizontal range of target domain scenes, f_r is the ground plane function of the target domain, and $\mu_\mathbf{t}$ and $\mathbf{\Sigma}_t$ are mean and covariance of the instance size (l, w, h) from the geometric statistics of the target domain, respectively. Finally, conflict detection is necessary to prevent overlapping between pseudobounding boxes.

2) Selecting Instances: Suitable instances in the source domain are filled into the pseudobounding boxes to form pseudoinstances. We define the viewing angle as the relative direction of the LiDAR sensor in the object canonical coordinate. The viewing angle α is a very important criterion for selecting suitable source domain instances to fill the pseudobounding box while maintaining the 2.5D characteristics of the LiDAR point clouds. The illustration of the viewing angle is shown in Fig. 2. The viewing angle α_2 of

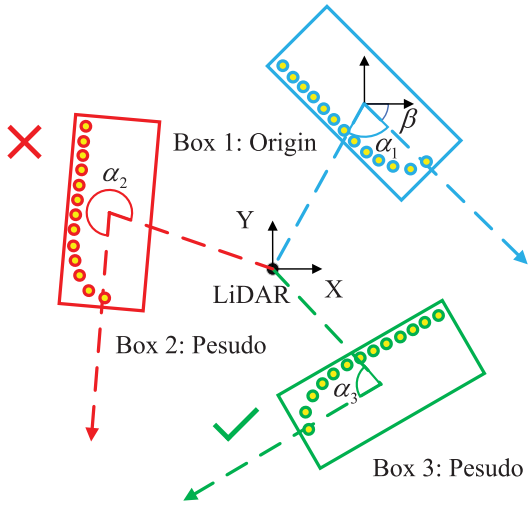


Fig. 2. Illustration of the viewing angle. Box 1 is the real point cloud instance, and Box 2 and Box 3 are the two pseudobounding boxes that are filled with the points transferred from Box 1. Among them, it is unreasonable that the points of Box 2 appear on the rear surface of the instance relative to the LiDAR sensor instead of the front surface. In contrast, Box 3 makes a reasonable transfer from Box 1 to fill the pseudobounding box. Therefore, the viewing angle α is an important criterion for selecting a suitable instance to be transferred.

the pseudobounding Box 2 does not match the viewing angle α_1 of the source domain instance bounding Box 1, so it is inappropriate to fill Box 2 with the points from Box 1 since the LiDAR should not see the rear surface in real scenes.

In contrast, Box 3 has a viewing angle α_3 close to the viewing angle α_1 of Box 1, which makes it sense to fill Box 3

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:32:53 UTC from IEEE Xplore. Restrictions apply.

Algorithm 1 Instance Transferring Method

Input

$\mathcal{P}_t$ is the point cloud scene in the target domain (with no instances);

$\mathcal{P}_s$ is an annotated point cloud in the source domain with the bounding boxes $\mathcal{B}_s$;

μ_t, $\mathbf{\Sigma}_t$ are the a priori geometric information of the target domain, i.e., the mean and covariance of the instance dimensions;

Distance range S and merging number M are the hyperparameters of this method;

Output

$\mathcal{P}_{trans}$ is the unadapted synthetic point cloud of the target domain;

$\mathcal{B}_{trans}$ is the pseudobounding boxes of $mathcal P_{trans}$;

1: Generate pseudobounding boxes $\mathcal{B}_{trans}$ in the target domain scene $\mathcal{P}_t$ as (2), based on its a priori geometric information μ_t and $\mathbf{\Sigma}_t$;
2: Remove the overlapping bounding boxes from $\mathcal{B}_{trans}$ (i.e., conflict detection);
3: Calculate the viewing angle α of all bounding boxes in $\mathcal{B}_s$ and all pseudobounding boxes $\mathcal{B}_{trans}$ according to (2), as $\mathcal{A}_s$ and $\mathcal{A}_{trans}$ respectively.
4: **for** each pseudobounding box $\mathbf{b}_i \in \mathcal{B}_{trans}$ and corresponding viewing angle $\alpha_i \in \mathcal{A}_{trans}$ **do**
5: Selecting M bounding boxes $\mathbf{b} \in \mathcal{B}_s$ that have similar viewing angles to α_i in distance range S as $\mathcal{B}_{candi}$;
6: Scale the point within $\mathbf{b} \in \mathcal{B}_{candi}$ in the source domain and fill into $\mathbf{b}_i$ in the target source scenes $\mathcal{P}_t$ as $\mathcal{P}_{trans}$.
7: **end for**
8: **return** $\mathcal{P}_{trans}$, $\mathcal{B}_{trans}$

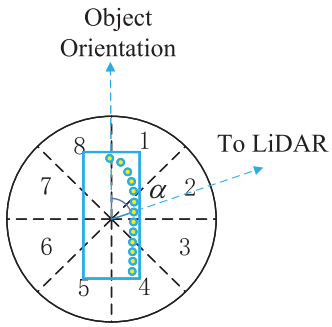


Fig. 3. Illustration of the viewing angle bins. We divide the continuous angle value into several bins to retrieve faster.

with the points in Box 1. Considering that the actual viewing angle is continuous value between $[0, 2\pi)$, we divide α into several bins as shown in Fig. 3 to retrieve instances with similar α faster. Given a bounding box in the source domain, the viewing angle α can be calculated as

$$\alpha = \arctan(v_2, v_1) \tag{3}$$

where

$$\mathbf{R} = \begin{bmatrix} \cos(\beta) & -\sin(\beta) & 0 \\ \sin(\beta) & \cos(\beta) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

and

$$\mathbf{v} = (v_1, v_2, v_3) = -\mathbf{R}^{-1} * \begin{bmatrix} x & y & z \end{bmatrix}$$

and

$$\mathbf{b} = \begin{bmatrix} x & y & z & l & w & h & \beta \end{bmatrix}$$

where $\mathbf{R}$ is the rotation matrix between the object canonical coordinate and the LiDAR coordinate, and $\mathbf{v}$ is the offset of the LiDAR sensor in the object canonical coordinate.

Besides, the qualities of the transferred instances are also crucial. Since the proposed data adaption method approximates the point cloud distribution of the target domain by modifying the point distribution of the source domain instances by adaptive downsampling (detail in Section III-D), the presented method needs to choose the instances with rich geometric information (redundant information) in the source domain, where the number of points (i.e., point cloud density) within an instance is used as a criterion of the integrity of geometric information it contains. Considering that the LiDAR point cloud density is inversely square to the distance, the proposed method takes the distance of the instance from the LiDAR as a simple criterion for measuring instance quality in this work. We set a distance range S as a hyperparameter to filter instances in the source domain. Meanwhile, objects of the same class generally have similar geometric structures. In this work, multiple instances from the source domain are merged by concatenation to achieve higher point density and more complete geometric information, which can alleviate the incomplete shape of some instances due to possible occlusions. Finally, the following adaptive downsampling method will remove the redundant points (detail in Section III-D). We randomly select M instances within the same viewing angle bin from the source domain instances, and then concatenate them directly as one instance after scale normalization. M and S will be discussed in detail in Section IV.

After picking the appropriate instances, we scale their size and fill them into the corresponding pseudobounding box. In summary, the instance transferring method is shown in Algorithm 1.

D. Adaptive Downsampling Method

The adaptive downsampling method aims to downsample the points of transferred instances to achieve an approximation to the point distribution in the target domain, as shown in Fig. 1.

1) Mapping Index: In this step, some prior knowledge of LiDAR is necessary to divide the continuous coordinates of the range view into discrete pixels of the range image, including the field of view (FOV) and angular resolution of LiDAR in the horizontal and vertical directions. These parameters can be obtained from the product manual of LiDAR given by the manufacturer (e.g., Velodyne HDL64E used in KITTI and Velodyne HDL32E used in nuScenes). To simplify the problem, we assume here that the vertical and horizontal resolutions (R_v and R_h, respectively) of LiDAR are consistent in the whole FOV. The range of the vertical and horizontal FOVs of LiDAR is limited in $[V_{\min}, V_{\max}]$,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:32:53 UTC from IEEE Xplore. Restrictions apply.

Algorithm 2 Adaptive Downsampling Method

Input

Unadaptive scenes $\mathcal{P}_{trans}$ and pseudobounding boxes $\mathcal{B}_{trans}$; Parameters about the LiDAR of the target domain V_{min}, V_{max}, H_{min}, H_{max}, R_v and R_h;

Output

$\mathcal{P}_a$ is the adapted synthetic point cloud of the target domain; $\mathcal{B}_a$ is the adapted pseudobounding boxes of $\mathcal{P}_a$;

1: Calculate the coordinates and index in the range view for each point $\mathbf{p} \in \mathcal{P}_{trans}$ according to (4) and (5);
2: Calculate the reprojection error for each point $\mathbf{p} \in \mathcal{P}_{trans}$ according to (10);
3: Calculate the indicator set $\mathcal{I}_a$, according to (11) and (12);
4: $\mathcal{P}_a = \{\mathbf{p} | ind = 1, (\mathbf{p}, ind) \in (\mathcal{P}_{trans}, \mathcal{I}_a)\}$
5: Get $\mathcal{B}_a$ by removing the empty pseudobounding boxes in $\mathcal{B}_{trans}$;
6: **return** $\mathcal{P}_a$, $\mathcal{B}_a$

$[H_{\min}, H_{\max}]$, respectively. For a point $\mathbf{p} = (x, y, z)$, the continuous coordinates (r, θ, ϕ) on the range view can be obtained by (4) and further quantized into discrete angular indexes (r, θ_d, ϕ_d) according to the parameters of the LiDAR, as shown in (5)

$$\begin{aligned} r &= \sqrt{x^2 + y^2 + z^2} \\ \theta &= \arccos\left(\frac{z}{r}\right) \\ \phi &= \arccos\left(\frac{y}{x}\right) \end{aligned} \tag{4}$$

$$\begin{aligned} \theta_d &= \left\lfloor \frac{\theta - V_{\min}}{R_v} \right\rfloor \\ \phi_d &= \left\lfloor \frac{\phi - H_{\min}}{R_h} \right\rfloor \end{aligned} \tag{5}$$

where $r, \theta, and \phi$ represent the depth (range), inclination, and azimuth in the range view, respectively.

2) Removing Redundant Points: LIDAR takes one measurement at each discrete angle making the points at each pixel in the range image unique. Using this uniqueness, we propose an adaptive downsampling method to remove the redundant points from the synthetic scene, which achieves an approximation to the real distribution of the target domain. To remove the redundant points by filtering all the same indexes in $\mathcal{P}_{\text{trans}}$, the set of corresponding points $\mathcal{R}_{(i,j)}$ on the pixel (i, j) in the range image can be expressed as (6)

$$\mathcal{R}_{(i,j)} = \{\mathbf{p} | \theta_d = i, \phi_d = j, \mathbf{p} \in \mathcal{P}_{\text{trans}}\} \tag{6}$$

where $\mathbf{p} = (x, y, z, r, \theta, \phi, \theta_d, \phi_d)$. Considering the occlusion relationship, the point with the closest distance on the same pixel will block the other points. For the point set $\mathcal{R}$ at each discrete angle (θ_d, ϕ_d), we only keep the closest point $\mathbf{p}$ as the sampling point. Therefore, the downsampling of the entire point set P_{trans} can be expressed as (7)

$$\mathcal{I}_b = \left\{ I(\mathbf{p} = \min_r \ \mathbf{p}_i \in \mathcal{R}_{(\theta_d, \phi_d)}) \, | \mathbf{p} \in \mathcal{P}_{\text{trans}} \right\} \tag{7}$$

where $\mathcal{I} = \{\text{ind}_1, \text{ind}_2, \ldots\}$ is an indicator set that $\text{ind}_i = 1$ indicates if point $\mathbf{p}_i$ is sampled, and I is an indicator function that $I(\text{True}) = 1$, $I(\text{False}) = 0$.

However, the complete foreground information is more important than the background information for the training of 3-D detectors. When the foreground point is obscured by a background point, we ensure the integrity of the foreground information by removing the background point. Similar to (6), the set of foreground points of at position (i, j) of the range image, $\mathcal{R}_f(i, j)$, can be obtained as (8)

$$\mathcal{R}_{f(i,j)} = \{\mathbf{p} | \theta_d = i, \phi_d = j, l = 1, (\mathbf{p}, l) \in (\mathcal{P}_{\text{trans}}, \mathcal{L}_{\text{trans}})\} \tag{8}$$

where $\mathcal{L}_{\text{trans}}$ is an indicator set to indicate whether the points in the $\mathcal{P}_{\text{trans}}$ come from the foreground (the transferred instance, $l = 1$) or the background (the target domain scene, $l = 0$). In combination with (7)–(9), the indicator set of the foreground-first downsampling $\mathcal{I}_f$ can be expressed as (9)

$$\begin{aligned} \mathcal{I}_f = \Big\{ & I(\mathbf{p} = \min_r \ \mathbf{p}_i \in \mathcal{R}_{(\theta_d, \phi_d)}) * I(\mathcal{R}_{f(\theta_d, \phi_d)} = \emptyset) \\ & + I(\mathbf{p} = \min_r \ \mathbf{p}_i \in \mathcal{R}_{f(\theta_d, \phi_d)}) * I(\mathcal{R}_{f(\theta_d, \phi_d)} \neq \emptyset) \\ & | \, \mathbf{p} \in \mathcal{P}_{\text{trans}} \Big\}. \end{aligned} \tag{9}$$

Note that (5) introduces quantization errors in the process of angle quantization. The quantization errors may lead to local point cloud density anomalies after downsampling, which deviates from the point distribution of the target domain LiDAR. We set a reprojection error threshold e to filter the point cloud to keep the sampled points from deviating too much from the sampling angle of the target domain LiDAR. The reprojection error threshold is set to a fixed value $e = 0.2$ m to ensure a reasonable quality of the synthesized point cloud. The reprojection error E is defined as (10)

$$E(\mathbf{p}) = \text{Norm}(x - x_r, y - y_r, z - z_r) \tag{10}$$

where

$$\begin{aligned} x_r &= r \cos(\theta_r) * \cos(\phi_r) \\ y_r &= r \cos(\theta_r) * \sin(\phi_r) \\ z_r &= r \sin \theta_r \end{aligned}$$

and

$$\begin{aligned} \theta_r &= (\theta_d + 0.5) * R_v + V_{\min} \\ \phi_r &= (\phi_d + 0.5) * R_h + H_{\min}. \end{aligned}$$

Since the point cloud scene is from the target domain, we only need to constrain the reprojection error of the foreground points. Similar to (8), the set of foreground points of at position (i, j) of the range image after filtered by reprojection error e, $\mathcal{R}_{\text{ef}(i,j)}$ can be obtained as (11)

$$\begin{aligned} \mathcal{R}_{\text{ef}(i,j)} = \{\mathbf{p} | \theta_d = i, \phi_d = j, l = 1, \\ E(\mathbf{p}) < e, (\mathbf{p}, l) \in (\mathcal{P}_{\text{trans}}, \mathcal{L}_{\text{trans}})\}. \end{aligned} \tag{11}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:32:53 UTC from IEEE Xplore. Restrictions apply.

Finally, the **adaptive downsampling method** can be described as (12)

$$\mathcal{I}_a = \Big\{ I(\mathbf{p} = \min_r \mathbf{p}_i \in \mathcal{R}_{(\theta_d,\phi_d)}) * I(\mathcal{R}_{\text{ef}(\theta_d,\phi_d)} = \emptyset) + I(\mathbf{p} = \min_r \mathbf{p}_i \in \mathcal{R}_{\text{ef}(\theta_d,\phi_d)}) * I(\mathcal{R}_{\text{ef}(\theta_d,\phi_d)} \neq \emptyset) \mid \mathbf{p} \in \mathcal{P}_{\text{trans}} \Big\}. \tag{12}$$

We discard redundant points from $\mathcal{P}_{\text{trans}}$ according to the indicators in $\mathcal{I}_a$. The adaptive downsampling method is shown as Algorithm 2.

IV. Experiments

A. Datasets and Evaluation Metrics

We follow [16], [17] to evaluate our data adaption method on various source–target combinations with the following datasets.

KITTI [1] is a popular autonomous driving dataset that consists of 3712 training samples and 3769 validation samples. The 3-D bounding box annotations are only provided for objects within the FoV of the front camera. Therefore, points outside of the FoV are ignored during training and evaluation. The objects are categorized into three levels (Easy, Moderate, and Hard) for evaluation according to the occlusion, truncation, and 2-D bounding box size. The evaluation metric of the KITTI is 3-D average precision (AP) under 40 recall thresholds (R40) and 11 recall thresholds (R11). The same as the previous works [16], [17], we report the results on the validation split using a 3-D AP under 11 recall thresholds when the target domain is the KITTI. The IoU threshold in this metric is 0.7.

Waymo Open Dataset [3] provides annotations for objects in the full 360°. There are 798 training sequences with around 158*k* LiDAR samples for training and 150 test sequences with around 30*k* LiDAR samples for evaluation. We subsample 1/10 of the training and validation sets. The Waymo Open Dataset uses high-quality LiDAR (64 beams) and has large-scale annotation data. The official 3-D detection evaluation metrics are mean average precision (mAP) (L1), mAP (L2), mAP with heading (mAPH) (L1), and mAPH (L2), where L1 and L2 denote the detection difficulty level. The L1 level includes the objects with more than five points and the L2 level includes the objects with more than one point. The mAPH metric takes into account object heading accuracy. The IoU thresholds in this metric are 0.7 vehicles (corresponds to cars in the KITTI and nuScenes dataset).

nuScenes Dataset [2] consists of 28 130 training samples and 6019 validation samples. The LiDAR used by nuScenes is Velodyne HDL32E, which has a lower detection range and point density than the first two. According to the nuScenes metrics, we adopt the three types of true positive metrics (TP metrics), including average translation error (ATE), average scale error (ASE), and average orientation error (AOE) for measuring translation scale, orientation errors from the nuScenes metrics, respectively. The commonly used mAP based on 3-D IoU with a threshold of 0.7 is adopted here to reflect the overall detection accuracy.

An overview of the dataset is shown in Table I. It can be seen that the three LiDAR have large gaps in performance and parameters (e.g., the number of beams, angular resolution, and point density), which makes them suitable for validating the effectiveness of our data adaption method approach across the different datasets.

B. Experimental Setup

1) Domain Pairs and Detectors: Without loss of generality, we use four pairs of source–target combinations separately to verify the effectiveness of our algorithm, namely, Waymo to KITTI, Waymo to nuScenes, nuScenes to KITTI, and KITTI to Waymo. These three pairs of combinations correspond to the adaption of different LiDAR data quality, namely, the adaption between the data with similar resolutions (Waymo to KITTI and KITTI to Waymo), the adaption of resolution from high to low (Waymo to nuScenes), and the adaption from low to high (nuScenes to KITTI). Since all the instances (i.e., positive samples) of the proposed data adaption method come from the annotation of the source domain dataset, the scales of the source domain datasets used here are larger than the target domain datasets except KITTI to Waymo task.

To verify the training effect of the synthetic scenes generated by the data adaption method on the target domain, we remove all the real instances of interest from the target domain scene during training. We only use the pseudoinstances to train the detector. This means that the detector does not see any real instances in the target domain during the whole training process, and the performance of the detector only depends on the quality of the synthetic instances built by the proposed data adaption method entirely.

Our experiments use the popular 3-D detector SECOND [7] for comparison to previous work. In addition, we evaluate our method on different 3-D detectors (i.e., PointRCNN [11], PointPillar [6], and PV-RCNN [12]) and compare with the replicated results of [16] to further validate the generalizability of our method.

2) Input Parameters: When the target domain is the KITTI, the detection range is set to be [0, 70.4 m] for the X-axis, [−40, 40 m] for the Y-axis, and [−3, 1 m] for the Z-axis. The voxel size is set to (0.05, 0.05, 0.1 m). When the target domain is the nuScenes dataset, the detection range is set to be [−51.2, 51.2 m] for both the X-axis and the Y-axis, and [−5, 3 m] for the Z-axis. The voxel size is set to (0.1, 0.1, 0.2 m) due to the large range of input data. When the target domain is the Waymo dataset, the detection range is set to be [0, 51.2 m] for the X-axis, [−51.2, 51.2 m] for the Y-axis, and [−2, 4 m] for the Z-axis. The voxel size is set to (0.1, 0.1, 0.15 m). The configs are consistent with [17] for fair comparison.

3) Implementation Details: Our experiments on four source–target domain pairs are based on the popular 3-D detector SECOND [7]. To further verify the detector-agnostic ability of the proposed method, four different types of detectors are used in the experiments of Waymo to KITTI task, including voxel-based detector SECOND [7], pillar-based detector PointPillar [6], point-based detector PointRCNN [11], and voxel-point-based detector PV-RCNN [12]. We adopt the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:32:53 UTC from IEEE Xplore. Restrictions apply.

TABLE II
PERFORMANCE OF THE PROPOSED DATA ADAPTION METHOD ON TWO SOURCE–TARGET PAIRS IN COMPARISON TO VARIOUS BASELINES AND SOTA METHODS. OUR DATA ADAPTION METHOD OUTPERFORMS ALL BASELINES AND SOTA METHODS. * MEANS METHODS THAT USE TARGET DOMAIN ANNOTATION INFORMATION

Waymo → KITTI				**nuScenes → KITTI**			
Method	Easy	Mod	Hard	Method	Easy	Mod	Hard
Direct Transfer	20.20	19.18	16.97	Direct Transfer	49.13	39.56	35.51
Wide-Range Aug	30.23	31.49	32.85	Wide-Range Aug	58.71	45.37	43.03
GT-AUG [7]	5.20	5.26	5.68	GT-AUG [7]	1.52	1.32	1.11
OT [16]*	39.78	37.82	39.55	OT [16]*	23.13	27.26	29.10
SN [16]	65.06	56.74	53.86	SN [16]	44.81	45.15	47.60
MLC-Net [17]	69.35	59.44	56.29	MLC-Net [17]	71.26	55.42	48.99
Ours	**70.83**	**62.09**	**62.24**	Ours	**76.95**	**63.29**	**61.78**
improvement	**+1.48**	**+2.65**	**+5.95**	*improvement*	**+5.69**	**+7.87**	**+12.79**
Oracle	87.99	78.32	77.11	Oracle	87.99	78.32	77.11

training settings of the popular point cloud detection (PCD) codebase OpenPCDet [54] to train our detectors using the pseudoinstances in the target domain scenes.

In the setting of hyperparameters, S, M, and pseudoinstance number N are determined according to the point cloud quality of the source and target domains. When the source domain is Waymo and the target domain is KITTI, we set $S = (20, 30]$, $M = 2$, $N = 8$. When the source domain is Waymo and the target domain is nuScenes, we set $S = (0, 10]$, $M = 1$, $N = 15$. When the source domain is nuScenes and the target domain is KITTI, we set $S = (0, 10]$, $M = 3$, $N = 8$. When the source domain is KITTI and the target domain is Waymo, we set $S = (3, 30]$, $M = 3$, $N = 15$.

4) Training and Inference Details: All the following experiments are performed on the Ubuntu 20.04 operating system with eight pieces of graphics processing units (GPUs) of NVIDIA RTX 3090 24 GB and dual Intel Xeon Gold 5218 CPU processor with 2.30-GHz frequency. The network training and testing process are implemented using Python 3.8 language environment and PyTorch 1.8.1.

We use Adam [55] with a learning rate of 0.0015 and weight decay of 0.01, and one cycle scheduler is used to train the detectors for 80 epochs, 30 epochs, and 30 epochs when the target domain is the KITTI, nuScenes, and Waymo, respectively. The batch size is set to be 16. We adopt commonly used data AUG strategies which include global random flipping about the X-axis or Y-axis, global scaling with a random scaling factor between 0.95 and 1.05, and global rotation about the Z-axis with a random angle factor between $-(1/4)\pi$ and $(1/4)\pi$. All the detectors are trained based on synthetic scenes with backgrounds from the target domain training set and foregrounds from instances in the source domain training set. Finally, the models are evaluated on the validation set of the target domain dataset.

For inference, we keep the top-100 proposals generated from voxel or point backbone for the two-stage detectors [11], [12] with a 3-D IoU threshold of 0.7 for nonmaximum suppression (NMS). For postprocessing, we use an NMS threshold of 0.1 to remove redundant boxes.

TABLE III
PERFORMANCE OF THE PROPOSED DATA ADAPTION METHOD ON WAYMO → NUSCENES TASK

Waymo → nuScenes				
Method	ATE	AST	AOE	AP_{3D}
Direct Transfer	0.27	0.23	**0.35**	17.42
Wide-Range Aug	0.27	0.22	**0.35**	17.90
SN [16]	0.27	0.20	**0.35**	18.57
GT-AUG [7]	**0.24**	**0.18**	0.36	1.23
Ours	0.31	0.21	**0.37**	**23.49**
Oracle	0.25	0.20	0.25	34.87

C. Benchmarking Result

We compare our method with the SOTA methods [16], [17] and baseline methods. Direct Transfer is a baseline method that directly evaluates the source domain pretrained model on the target domain, and Oracle indicates the fully supervised model trained on the target domain. Wide-Range Aug is another baseline method with random scaling AUG of a wide range which potentially includes the target domain scales.

The performance of the proposed data adaption method on four source–target domain pairs is shown in Tables II–IV, where our method outperforms all previous works by a large margin. On the Waymo → KITTI task, the proposed method achieves AP improvements of $+$**1.48%**, $+$**2.65%**, and $+$**5.95%** on *Easy*, *Moderate*, and *Hard*, respectively, compared

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:32:53 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
PERFORMANCE OF THE PROPOSED DATA ADAPTION METHOD ON KITTI → WAYMO TASK

KITTI → Waymo				
Methods	AP/L1	APH/L1	AP/L2	APH/L2
Direct Transfer	9.17	8.99	7.94	7.78
Wide-Range Aug	18.61	18.18	16.77	16.40
OT [16]	26.48	25.84	23.85	23.29
SN [16]	30.69	30.06	27.23	26.67
MLC-Net [17]	38.21	37.74	34.46	34.04
GT-AUG [7]	3.12	3.01	2.88	2.63
Ours	**40.32**	**39.78**	**36.11**	**35.94**
Oracle	70.96	70.34	62.58	62.02

with the SOTA method [17]. Similarly, on the nuScenes → KITTI task, the proposed methods improve the AP of the existing methods by +**5.69%**, +**7.87%**, and +**12.79%** on the three levels, respectively. On the Waymo → nuScenes task, our method still improves the performance by +**4.92%** compared with the existing methods. On the KITTI → Waymo task, our method improves the performance by +2.04% and + 1.90% compared with the existing methods. It should be noted that there are about 6.1M annotated vehicles in the Waymo dataset. Even after 1/10 sampled, this number is still far more than the number of annotated cars in KITTI (about 20k). In the proposed method, the pseudoinstances which are used as the positive samples during training are transferred from the source domain annotation. Thus, an overfitting problem is inevitable in the KITTI to the Waymo task, which is one of the reasons why its performance is far inferior compared with the performance of the Waymo → KITTI task.

In addition, to further validate the generalization of the proposed method, we evaluate the performance of our data adaption method on several popular 3-D detectors, as shown in Table V. The evaluation results show that our method yields consistent and significant performance improvements on different 3-D detectors, which illustrates that the proposed method is model-agnostic and can be applied to any 3-D detector without modification. On four different types of popular 3-D detectors [6], [7], [11], [12], our proposed method achieves an average performance improvement of +**5.87%**, +**4.80%**, and +**5.88%** on *Easy*, *Moderate*, and *Hard*, respectively.

It is worth noting that our method is actually trained directly on the target domain with synthetic data, and we only use the annotations of the source domain without training on the source domain, unlike all previous methods. This means that our model does not need any pretraining and can be deployed more flexibly and quickly than other unsupervised methods. Meanwhile, in our experiments, we remove all real instances in the target domain which means that the detectors cannot actually see any real instances in the target domain during the entire training, which makes the hyperparameters in this method have a significant impact on the model performance.

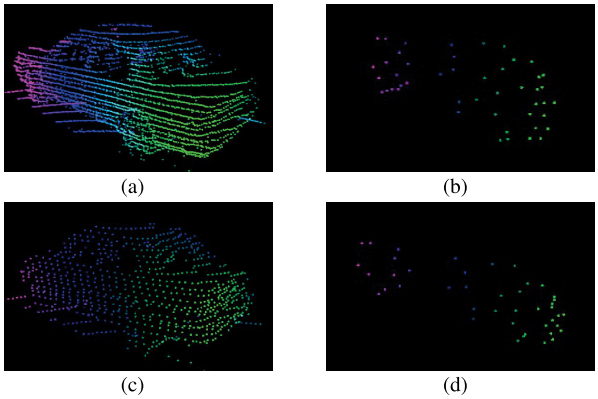


Fig. 4. Quality of pseudoinstances in the target domain is affected by the quality of instances in the source domain. (a) and (c) Two instances with different qualities from the source domain, which are synthesized into the same position in the target domain as shown in (b) and (d). By adaptive downsampling, a distribution that approximates the target domain is obtained in (b), while (d) cannot be well approximated by adaptive downsampling due to fewer points.

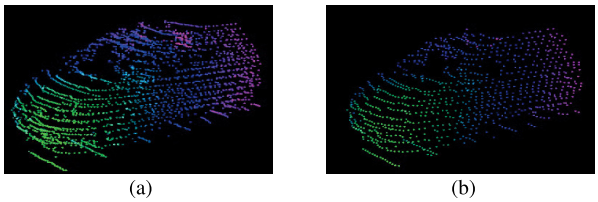


Fig. 5. Visualization of the merged instance and the corresponding synthesized pseudoinstances. (a) Car instance 3 obtained by merging three instances in the source domain in the range of 20–30 m. (b) Synthesize car instance 3 to a location 15 m away from LiDAR in the KITTI scene.

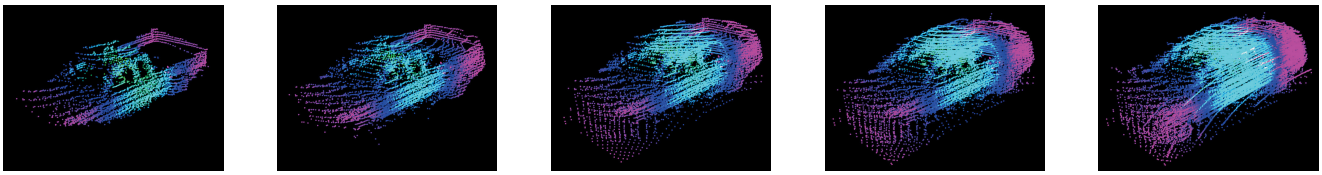
D. Discussion

In this section, we conduct several experiments on the hyperparameters involved in our data adaption method to evaluate their impacts: *distance range* S, *merge number* M, and *pseudoinstance number* N.

1) Distance Range S: We conduct several experiments to study the impact of the hyperparameter S, which is used to specify the distance interval for sampling instance in the source domain instance: the closer the instance is to the LiDAR, the denser the points on its surface, the less likely it to be occluded, which means the more complete geometric and higher quality of instances; for the instances that are far away from the LiDAR, their surfaces may be covered by fewer points and are easily occluded which means losing a lot of geometric information, resulting in low-quality object instances.

The results are shown in Table VI. It can be seen that when $S = [20, 30)$, it achieves the best results. As the sampling range becomes further, the performance on *Easy* decreases monotonically as the quality of the point cloud decreases, so does the quality of the synthesized samples at close range. The performance of *Mod* and *Hard* increases first and then decreases as the sampling distance increases. We consider that this may be caused by two reasons: 1) the larger number of samples improves the generalization ability of the network and 2) due to the nonuniform beam distribution of the LiDAR

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:32:53 UTC from IEEE Xplore. Restrictions apply.



(a) (b) (c) (d) (e)

Fig. 6. Qualitative visualization of the influence of merge number M on geometric information of pseudoinstance. (a) $M = 1$. (b) $M = 2$. (c) $M = 3$. (d) $M = 4$. (e) $M = 5$.

TABLE V

PERFORMANCE OF OUR DATA ADAPTION METHOD ON DIFFERENT POPULAR 3-D DETECTORS. THE RESULTS SHOW THAT OUR METHOD HAS A CONSISTENT PERFORMANCE BOOST ON DIFFERENT DETECTORS COMPARED WITH THE SOTA METHOD [16]. *: THE METHOD IS REPLICATED BY US ON THE CORRESPONDING MODEL

Waymo → KITTI					
Detector	Type	Method	Easy	Mod	Hard
SECOND [7]	Voxel-based	Source only	20.20	19.18	16.97
		SN [16]*	65.06	56.74	53.86
		Ours	70.83	62.09	62.24
		imporovement	**+5.77**	**+5.35**	**+8.38**
		Oracle	87.99	78.32	77.11
PointPillar [6]	Pillar-based	Source only	16.82	17.06	16.14
		SN [16]*	66.54	54.85	52.90
		Ours	70.09	59.52	59.37
		imporovement	**+3.55**	**+4.67**	**+6.47**
		Oracle	86.15	74.21	67.97
PointRCNN [11]	Point-based	Source only	20.22	21.43	20.49
		SN [16]*	61.93	58.07	58.44
		Ours	72.85	62.88	61.09
		imporovement	**+10.92**	**+4.81**	**+2.65**
		Oracle	88.93	78.16	70.42
PV-RCNN [12]	Point-Voxel-based	Source only	25.42	24.96	23.95
		SN [16]*	72.59	61.01	57.58
		Ours	75.83	65.42	63.60
		imporovement	**+3.25**	**+4.41**	**+6.02**
		Oracle	89.16	83.43	78.65
Average Performance Improvement			**+5.87**	**+4.80**	**+5.88**

used by Waymo, the density drop of the point cloud is not significant at moderate distances. To verify our hypothesis, we perform three experiments using $S = [0, 30)$, $S = [10, 30)$, and $S = [10, 40)$. These are essentially equivalent to the performance tradeoff obtained with the previous hyperparameter values. It also greatly improves the performance of *Mod* and *Hard* compared with $S = [0, 10)$ and $S = [10, 20)$ while maintaining the performance on *Easy* as shown in Table VI.

Since our method can only perform adaptive downsampling to the transferred instances, it is necessary to ensure the number of points in the sampled instances; otherwise, the point cloud synthesized at close range will not be able to realistically simulate the distribution of the target domain because the limited points' qualitative visualization is shown in Fig. 4.

2) Merge Number M: This hyperparameter is also used to control the quality of the instances to be transferred, which is a tradeoff between the precise and integrity of geometric information. The results of experiments with different M are shown in Table VII. The results show that when $M = 2$, our data adaption method has the best performance.

Since our data adaption method modifies the distribution of points through adaptive downsampling, the instances with far distances will not able to obtain point clouds as dense as real close objects if we transfer distant object instances from the source domain into a closer position to the target domain scene, which is limited by the number of points. When $M = 1$, the performance of the synthesized data in close range decreases significantly, which is due to the lack of points resulting in the distribution deviation in close range. After the points are concatenated, the instance will have a multilayer contour, as shown in Fig. 5. However, the proposed adaptive downsampling method will remove redundant points and only the outermost contour of the instance will be retained, and the contour of the object is determined by the most

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:32:53 UTC from IEEE Xplore. Restrictions apply.

TABLE VI
ANALYSIS ON THE DISTANCE RANGE S ON KITTI VALIDATION SPLITS WITH $m = 2$ AND $n = 8$

Waymo → KITTI				
S	Easy	Mod	Hard	Sample num
$[0, 10)$	**73.85**	55.80	55.81	67533
$[10, 20)$	73.73	57.11	56.71	136929
$[20, 30)$	70.83	**62.09**	**62.24**	170658
$[30, 40)$	66.52	58.47	57.94	190324
$[40, 50)$	58.69	54.38	51.89	184587
$[0, 30)$	73.11	59.89	57.33	375120
$[10, 30)$	71.88	58.41	55.98	307587
$[10, 40)$	68.41	59.42	58.32	497911

TABLE VII
ANALYSIS ON THE MERGING NUMBER OF INSTANCE M ON KITTI VALIDATION SPLITS WHEN $S = [20, 30)$ AND $N = 8$

Waymo → KITTI			
M	Easy	Mod	Hard
1	62.34	52.40	52.79
2	70.83	**62.09**	**62.24**
3	**75.59**	61.29	58.27
4	71.30	60.29	59.72
5	70.99	58.05	56.79

prominent part of the concatenated point cloud contours. When the number of merges increases, the inserted instance will have more dense point clouds, but its geometry will deviate from the real situation more and more. When $M > 2$, the performance of the final training model decreases gradually due to the deviation of geometry information. The qualitative visualization of a geometric information deviation is shown in Fig. 6. When $M = 1$, the pseudoinstance has the most accurate geometric information (a pick-up truck), but part of its point cloud is obscured. When $M = 2$ or 3, by merging multiple point clouds, the point cloud information becomes more complete, while the details on the tail are blurred. When $M = 5$, the information of the car body is completely blurred, which results in that most of the pseudoinstances have similar geometric shapes, similar to a van.

3) Pseudoinstance Num N: This hyperparameter controls the maximum number of instances transferred into the same target domain scene. Compared with the first two parameters, the meaning of this parameter is more intuitive. Smaller N only provides a few positive samples for the detector to learn. On the contrary, considering the occlusion relation between instances, the distant instances are more likely to be occluded when there are too many instances in the same scene, which will reduce the quality of positive samples. Several experiments on this parameter are shown in Table VIII, and it can be seen that the model can obtain the best performance with $N = 8$. The qualitative visualization is shown in Fig. 7. The instances in Fig. 7(a) are too dense resulting in a large amount of occlusion, while the more complete geometric information of the instances is preserved in Fig. 7(b).

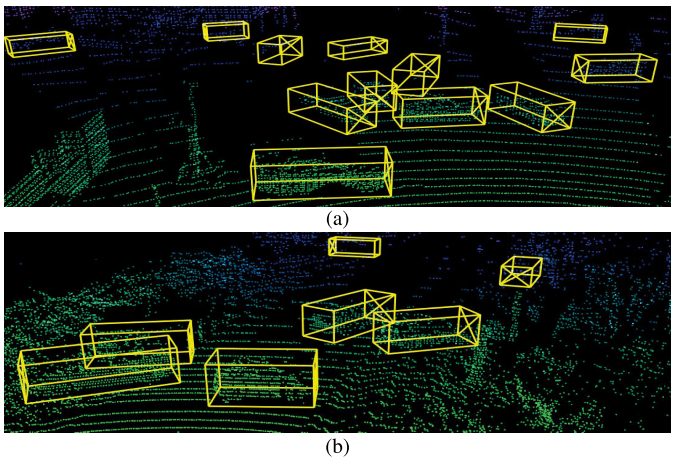


Fig. 7. Visualization of the synthesized scenes with different pseudoinstances numbers. (a) Too many synthetic instances cause severe occlusion. (b) Appropriate number of synthetic instances.

TABLE VIII
ANALYSIS ON THE NUMBER OF PSEUDOINSTANCES N ON THE KITTI VALIDATION DATASET WHEN $S = [20, 30)$ AND $M = 2$

Waymo → KITTI			
N	Easy	Mod	Hard
4	70.05	57.31	56.06
8	70.83	**62.09**	**62.24**
12	**70.86**	57.87	56.08
16	70.17	55.58	55.74

E. Qualitative Visualization

Visualization of our synthetic scenes is shown in Fig. 8. The point distribution of our pseudoinstances is very close to the real instances in the target domain, i.e., data adaptation. The detector is trained by the synthetic point clouds and pseudoinstances to achieve adaptation in the target domain.

V. FUTURE WORK

In this article, we use the scenes with no interest instances in the target domain to better evaluate the training effect of the synthetic scenes generated by our data adaption method. Therefore, the distribution similarity between the synthetic point cloud and the target domain dominates the performance of the detectors, which makes the performances more sensitive to the hyperparameters in the data adaption method. At the same time, scenes with no interest instances may not be

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:32:53 UTC from IEEE Xplore. Restrictions apply.

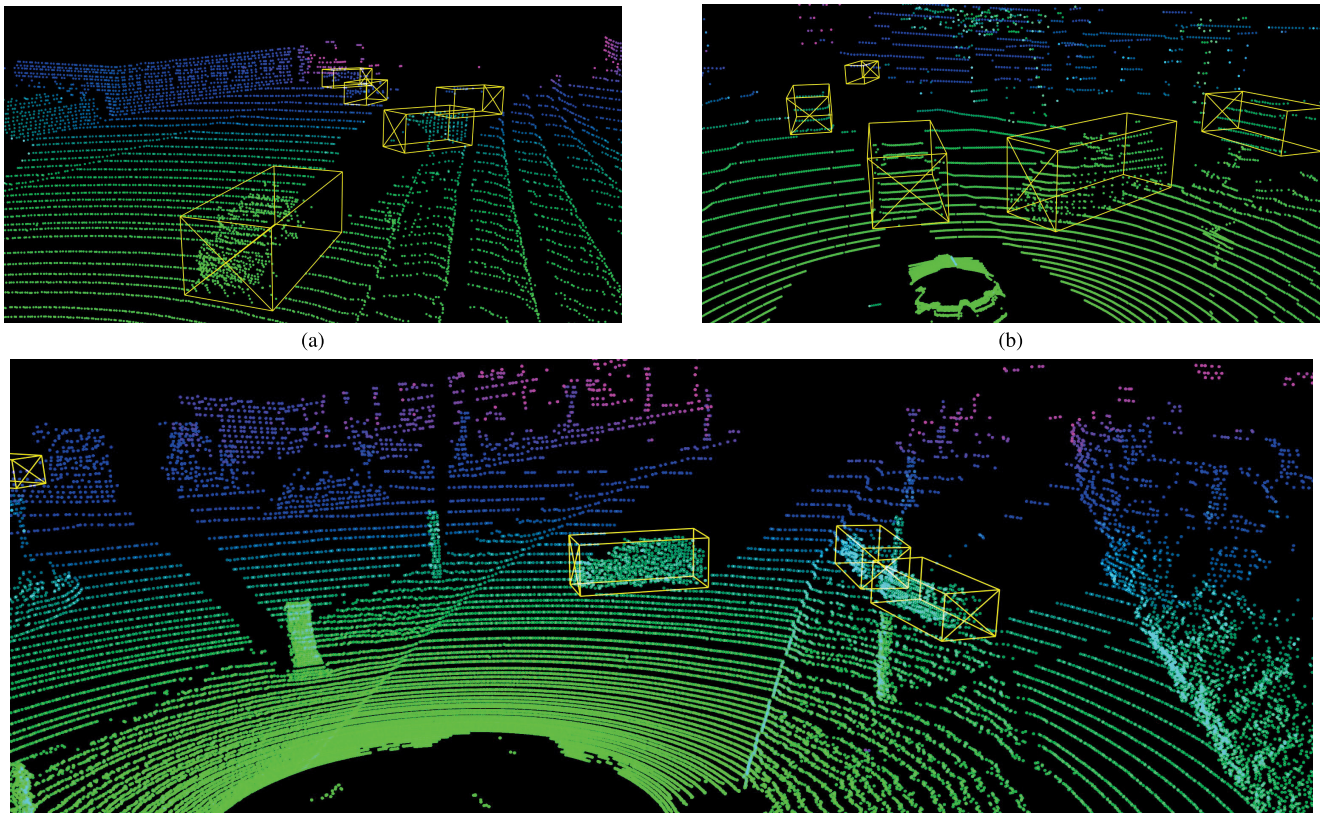


Fig. 8. Point cloud scene with pseudoinstances synthesized by the proposed data adaption method. Our data adaption method is able to transfer the existing point cloud instances into any scene collected by any LiDAR. The synthesized pseudoinstances are very similar to the real instance in data obtained by LiDAR scanning, and its surface point cloud distribution is determined according to the LiDAR sensor parameters and distance while keeping a reasonable occlusion relationship between instances and background. (a) Synthesized pseudoinstances in the Waymo → KITTI task. (b) Synthesized pseudoinstances in the Waymo → nuScenes task. (c) Qualitative visualization of synthesized scene in the Waymo → KITTI task.

available in the general scenarios. We think it might be a good idea to extend our data adaption method to pseudolabel-based unsupervised learning (i.e., self-training). Since the proposed data adaption method can synthesize pseudoinstances in the scenes, it is equivalent to the situation that a portion of the annotations of the input data are available when there are real instances in the target domain (i.e., miss annotated). This is similar to the pseudolabel-based method, except that our pseudobounding boxes are absolutely accurate. In this way, trying to mine the instances of the target domain during the training process may further improve the performance of the model and improve the robustness of training. We will continue to research this aspect in the future.

VI. Conclusion

In this article, we propose a data adaption method to address the unsupervised domain adaption problem for 3-D detection, which trains the model by transferring the source domain objects with adaptive point distribution into the target domain scenes. To the best of our knowledge, the proposed method is the first to solve the UDA problem from the perspective of data and can be applied to any 3-D detector. Experiments show that our method achieves SOTA performance on source–target pairs across multiple datasets without any real instances of the target domain. The experiments also show that the proposed method has consistent performance improvements on multiple popular 3-D detectors.

References

[1] A. Geiger, P. Lenz, C. Stiller, and R. Urtasun, "Vision meets robotics: The KITTI dataset," *Int. J. Robot. Res.*, vol. 32, no. 11, pp. 1231–1237, 2013.

[2] H. Caesar et al., "NuScenes: A multimodal dataset for autonomous driving," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2020, pp. 11621–11631.

[3] P. Sun et al., "Scalability in perception for autonomous driving: Waymo open dataset," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2020, pp. 2446–2454.

[4] Y. Zhou and O. Tuzel, "VoxelNet: End-to-end learning for point cloud based 3D object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 4490–4499.

[5] J. Deng, S. Shi, P. Li, W. Zhou, Y. Zhang, and H. Li, "Voxel R-CNN: Towards high performance voxel-based 3D object detection," 2020, *arXiv:2012.15712*.

[6] A. H. Lang, S. Vora, H. Caesar, L. Zhou, J. Yang, and O. Beijbom, "Pointpillars: Fast encoders for object detection from point clouds," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2019, pp. 12697–12705.

[7] Y. Yan, Y. Mao, and B. Li, "SECOND: Sparsely embedded convolutional detection," *Sensors*, vol. 18, no. 10, p. 3337, 2018.

[8] W. Zheng, W. Tang, S. Chen, L. Jiang, and C.-W. Fu, "CIA-SSD: Confident IoU-aware single-stage object detector from point cloud," 2020, *arXiv:2012.03015*.

[9] J. Mao et al., "Voxel transformer for 3D object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2021, pp. 3164–3173.

[10] Z. Li, F. Wang, and N. Wang, "LiDAR R-CNN: An efficient and universal 3D object detector," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 7546–7555.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:32:53 UTC from IEEE Xplore. Restrictions apply.

[11] S. Shi, X. Wang, and H. Li, "PointRCNN: 3D object proposal generation and detection from point cloud," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 770–779.

[12] S. Shi et al., "PV-RCNN: Point-voxel feature set abstraction for 3D object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2020, pp. 10529–10538.

[13] S. Shi et al., "PV-RCNN++: Point-voxel feature set abstraction with local vector representation for 3d object detection," 2021, *arXiv:2102.00463*.

[14] H. Wu, J. Deng, C. Wen, X. Li, C. Wang, and J. Li, "CasA: A cascade attention network for 3-D object detection from LiDAR point clouds," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–11, 2022.

[15] C. Yu, J. Lei, B. Peng, H. Shen, and Q. Huang, "SIEV-Net: A structure-information enhanced voxel network for 3D object detection from LiDAR point clouds," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–11, 2022.

[16] Y. Wang et al., "Train in Germany, test in the USA: Making 3D object detectors generalize," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2020, pp. 11713–11723.

[17] Z. Luo et al., "Unsupervised domain adaptive 3D detection with multi-level consistency," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2021, pp. 8866–8875.

[18] L. Fan et al., "Embracing single stride 3D object detector with sparse transformer," 2021, *arXiv:2112.06375*.

[19] T.-Y. Lin et al., "Microsoft COCO: Common objects in context," in *Proc. Eur. Conf. Comput. Vis.* New York, NY, USA: Springer, 2014, pp. 740–755.

[20] A. Gupta, P. Dollar, and R. Girshick, "LVIS: A dataset for large vocabulary instance segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 5356–5364.

[21] Y. Chen, W. Li, C. Sakaridis, D. Dai, and L. Van Gool, "Domain adaptive faster R-CNN for object detection in the wild," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 3339–3348.

[22] K. Saito, Y. Ushiku, T. Harada, and K. Saenko, "Strong-weak distribution alignment for adaptive object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 6956–6965.

[23] Y. Zheng, D. Huang, S. Liu, and Y. Wang, "Cross-domain object detection through coarse-to-fine feature adaptation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 13766–13775.

[24] D. Guan, J. Huang, A. Xiao, S. Lu, and Y. Cao, "Uncertainty-aware unsupervised domain adaptation in object detection," *IEEE Trans. Multimedia*, vol. 24, pp. 2502–2514, 2021.

[25] D. Guan, J. Huang, S. Lu, and A. Xiao, "Scale variance minimization for unsupervised domain adaptation in image segmentation," *Pattern Recognit.*, vol. 112, Apr. 2021, Art. no. 107764.

[26] J. Yang, S. Shi, Z. Wang, H. Li, and X. Qi, "ST3D: Self-training for unsupervised domain adaptation on 3D object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2021, pp. 10368–10378.

[27] C. R. Qi, H. Su, K. Mo, and L. J. Guibas, "PointNet: Deep learning on point sets for 3D classification and segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jul. 2017, pp. 652–660.

[28] C. R. Qi, L. Yi, H. Su, and L. J. Guibas, "PointNet++: Deep hierarchical feature learning on point sets in a metric space," 2017, *arXiv:1706.02413*.

[29] C. R. Qi, O. Litany, K. He, and L. J. Guibas, "Deep Hough voting for 3D object detection in point clouds," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2019, pp. 9277–9286.

[30] Z. Yang, Y. Sun, S. Liu, and J. Jia, "3DSSD: Point-based 3D single stage object detector," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2020, pp. 11040–11048.

[31] Y. Zhang, Q. Hu, G. Xu, Y. Ma, J. Wan, and Y. Guo, "Not all points are equal: Learning highly efficient point-based detectors for 3D LiDAR point clouds," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2022, pp. 18953–18962.

[32] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 28, 2015, pp. 91–99.

[33] W. Liu et al., "SSD: Single shot multibox detector," in *Proc. Eur. Conf. Comput. Vis.* New York, NY, USA: Springer, 2016, pp. 21–37.

[34] B. Graham and L. van der Maaten, "Submanifold sparse convolutional networks," 2017, *arXiv:1706.01307*.

[35] B. Liu, M. Wang, H. Foroosh, M. Tappen, and M. Pensky, "Sparse convolutional neural networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2015, pp. 806–814.

[36] T. Yin, X. Zhou, and P. Krahenbuhl, "Center-based 3D object detection and tracking," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 11784–11793.

[37] K. Duan, S. Bai, L. Xie, H. Qi, Q. Huang, and Q. Tian, "CenterNet: Keypoint triplets for object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 6569–6578.

[38] H. Law and J. Deng, "CornerNet: Detecting objects as paired keypoints," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 734–750.

[39] J. Noh, S. Lee, and B. Ham, "HVPR: Hybrid voxel-point representation for single-stage 3D object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2021, pp. 14605–14614.

[40] Y. Ganin and V. Lempitsky, "Unsupervised domain adaptation by backpropagation," in *Proc. Int. Conf. Mach. Learn.*, 2015, pp. 1180–1189.

[41] J. Hoffman, D. Wang, F. Yu, and T. Darrell, "FCNs in the wild: Pixel-level adversarial and constraint-based adaptation," 2016, *arXiv:1612.02649*.

[42] Y. Zhang, Z. Qiu, T. Yao, D. Liu, and T. Mei, "Fully convolutional adaptation networks for semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 6810–6818.

[43] J. Hoffman et al., "CyCADA: Cycle-consistent adversarial domain adaptation," in *Proc. Int. Conf. Mach. Learn.*, 2018, pp. 1989–1998.

[44] Q. Cai, Y. Pan, C.-W. Ngo, X. Tian, L. Duan, and T. Yao, "Exploring object relation in mean teacher for cross-domain detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 11457–11466.

[45] M. Khodabandeh, A. Vahdat, M. Ranjbar, and W. Macready, "A robust learning approach to domain adaptive object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 480–490.

[46] K. Saito, Y. Ushiku, and T. Harada, "Asymmetric tri-training for unsupervised domain adaptation," in *Proc. Int. Conf. Mach. Learn.*, 2017, pp. 2988–2997.

[47] Y. Zou, Z. Yu, B. Kumar, and J. Wang, "Unsupervised domain adaptation for semantic segmentation via class-balanced self-training," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 289–305.

[48] C. Qin, H. You, L. Wang, C.-C. J. Kuo, and Y. Fu, "PointDAN: A multi-scale 3D domain adaption network for point cloud representation," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 32, 2019, pp. 1–12.

[49] L. Yi, B. Gong, and T. Funkhouser, "Complete & label: A domain adaptation approach to semantic segmentation of LiDAR point clouds," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 15363–15373.

[50] M. Jaritz, T.-H. Vu, R. de Charette, E. Wirbel, and P. Perez, "XMUDA: Cross-modal unsupervised domain adaptation for 3D semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 12605–12614.

[51] B. Wu, A. Wan, X. Yue, and K. Keutzer, "SqueezeSeg: Convolutional neural nets with recurrent CRF for real-time road-object segmentation from 3D LiDAR point cloud," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, May 2018, pp. 1887–1893.

[52] J.-Y. Zhu, T. Park, P. Isola, and A. A. Efros, "Unpaired image-to-image translation using cycle-consistent adversarial networks," in *Proc. IEEE Int. Conf. Comput. Vis.*, Oct. 2017, pp. 2223–2232.

[53] N. Inoue, R. Furuta, T. Yamasaki, and K. Aizawa, "Cross-domain weakly-supervised object detection through progressive domain adaptation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 5001–5009.

[54] (2020). O. D. Team. *OpenPCDet: An Open-Source Toolbox for 3D Object Detection From Point Clouds*. [Online]. Available: https://github.com/open-mmlab/OpenPCDet

[55] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," 2014, *arXiv:1412.6980*.



Diankun Zhang received the B.S. degree from the Beijing University of Technology, Beijing, China, in 2019. He is pursuing the Ph.D. degree with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing.

His research interests include deep learning, point cloud processing, and object detection.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:32:53 UTC from IEEE Xplore. Restrictions apply.



Xueqing Wang received the B.Eng. degree from the University of Science and Technology, Beijing, China, in 2020. She is pursuing the M.D. degree with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing.

Her research interests include deep-learning-based signal processing and 3-D object detection.



Zhijie Zheng received the B.E. degree from the Huazhong University of Science and Technology, Wuhan, China, in 2019. He is pursuing the Ph.D. degree with the University of Chinese Academy of Sciences, Beijing, China, and Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing.

His research interests include perception beyond the visible spectrum, deep learning, and cross-modal learning.



Xiaojun Liu (Member, IEEE) received the B.S. and M.S. degrees in electronics science and technology from the North University of China, Taiyuan, China, in 1995 and 1998, respectively, and the Ph.D. degree in electrical engineering from the Chinese Academy of Sciences (CAS), Beijing, China, in 2001.

Since 2001, he has been with the Key Laboratory of Electromagnetic Radiation and Sensing Technology, CAS, where he is a Professor and currently leads a group on the researches of remote sensing of ice sheets with projects supported by National Natural Science Foundation of China, National High Technology Research and Development Projects (863 Projects) of China, etc. His research interests include application of radars to the remote sensing of the polar ice sheets, ionosphere, and land. He developed several radar systems currently being used at Institute of Electronics, Chinese Academy of Sciences (IECAS) for sounding and imaging of polar ice sheets. These radar systems have been used in field experiments in Antarctica for five times since CHINARE 26. In the field of lunar and deep space exploration, he leads a group to demonstrate the scheme of subsurface penetrating radar of Mars Orbiter and Vehicle for China's first Mars exploration. The program has been adopted and is being implemented.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:32:53 UTC from IEEE Xplore. Restrictions apply.